

Chapter

Factorization Of Polynomials

1. If $x^2 + kx + 6 = (x - 2)(x - 3)$ for all values of x , then the value of k is:

- | | |
|--------|--------|
| (a) -5 | (b) -3 |
| (c) -2 | (d) 5 |

SQP 2024

2. What must be subtracted from the polynomial $x^3 + x^2 - 2x + 1$, so that the result is exactly divisible by $(x - 3)$?

- | | |
|---------|---------|
| (a) -31 | (b) -30 |
| (c) 30 | (d) 31 |

Main 2024

3. If $x - 2$ is a factor of $x^3 - kx - 12$, then the value of k is:

- | | |
|--------|--------|
| (a) 3 | (b) 2 |
| (c) -2 | (d) -3 |

Main 2023

4. Factorize completely using factor theorem:

$$2x^3 - x^2 - 13x - 6$$

Main 2023

5. Find the value of a if $x - a$ is a factor of the polynomial $3x^3 + x^2 - ax - 81$

Main 2023

6. If a polynomial $x^3 + 3x^2 + 3x + 1$ is divided by $(x + 1)$, then the remainder is :

(a) -31

(b) 0

(c) 11

(d) -11

Comp 2023

7. If $p(x) = x^3 - ax^2 + bx + 3$ leaves a remainder -19 when divided by $(x + 2)$ and remainder 17 when divided by $(x - 2)$, prove that $a + b = 6$.

Comp 2023

8. $x^2 + 2x^2 - 5ax - 8$ and $x^3 + ax^2 - 12x - 6$ when divided by $(x - 2)$ and $(x - 3)$ leave remainder p and q respectively. If $q - p = 10$, find the value of a .

Comp 2023

9. Polynomial $x^2 - 2x^2 + ax + 12$ when divided by $(x + 1)$ leaves a remainder 20, then a is :

(a) -31

(b) 9

(c) 11

(d) -11

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10. If a polynomial $2x^2 - 7x - 1$ is divided by $x + 3$, then the remainder is :

(a) -4

(b) 38

(c) -3

(d) 2

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11. If $(x - 2)$ is a factor of the polynomial $x^3 + 2x^2 - 13x + k$, then k is :

(a) -10

(b) 26

(c) -26

(d) 10

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Solution

1.

Ans: (a) -5

Explanation:

Given,

$$x^2 + kx + 6 = (x - 2)(x - 3)$$

Expanding the right-hand side,

$$\begin{aligned}(x - 2)(x - 3) &= x(x - 3) - 2(x - 3) \\ &= x^2 - 3x - 2x + 6 \\ &= x^2 - 5x + 6\end{aligned}$$

So,

$$x^2 + kx + 6 = x^2 - 5x + 6$$

Comparing coefficients of x ,

$$k = -5$$

Hence, **option (a) is the correct answer.**

2.

Ans: (a) -31

Explanation:

Let k be the number to be subtracted from the polynomial

$$x^3 + x^2 - 2x + 1$$

Then,

$$p(x) = x^3 + x^2 - 2x + 1 + k$$

The result will be exactly divisible by $(x - 3)$ if

$$p(3) = 0$$

$$3^3 + 3^2 - 2(3) + 1 + k = 0$$

$$27 + 9 - 6 + 1 + k = 0$$

$$31 + k = 0$$

$$k = -31$$

Hence, **option (a)** is the correct answer.

3.

Ans: (c) -2

Explanation:

Given,

$$f(x) = x^3 - kx - 12$$

Since $(x - 2)$ is a factor of the given polynomial,

by the **Factor Theorem**,

$$f(2) = 0$$

Substituting $x = 2$,

$$2^3 - k(2) - 12 = 0$$

$$8 - 2k - 12 = 0$$

$$-2k = 4$$

$$k = -2$$

Hence, **option (c)** is the correct answer.

4.

Ans:

Let

$$p(x) = 2x^3 - x^2 - 13x - 6$$

We use the **hit and trial method**.

Substituting $x = -2$,

$$\begin{aligned} p(-2) &= 2(-2)^3 - (-2)^2 - 13(-2) - 6 \\ &= 2(-8) - 4 + 26 - 6 \\ &= -16 - 4 + 26 - 6 \\ &= 0 \end{aligned}$$

Hence, $x = -2$ or $(x + 2)$ is a factor of $p(x)$.

Dividing $p(x)$ by $(x + 2)$, we get the quotient

$$2x^2 - 5x - 3$$

So,

$$2x^3 - x^2 - 13x - 6 = (x + 2)(2x^2 - 5x - 3)$$

Now factorizing the quadratic factor,

$$\begin{aligned} 2x^2 - 5x - 3 &= 2x^2 - 6x + x - 3 \\ &= 2x(x - 3) + 1(x - 3) \\ &= (2x + 1)(x - 3) \end{aligned}$$

Hence,

$$2x^3 - x^2 - 13x - 6 = (x + 2)(2x + 1)(x - 3)$$

5.

Ans:

We have

$$p(x) = 3x^3 + x^2 - ax - 81$$

Since $(x - a)$ is a factor of $p(x)$, by the **Factor Theorem**,

$$p(a) = 0$$

Substituting $x = a$,

$$3a^3 + a^2 - a(a) - 81 = 0$$

$$3a^3 + a^2 - a^2 - 81 = 0$$

$$3a^3 - 81 = 0$$

$$3a^3 = 81$$

$$a^3 = 27$$

$$a = \sqrt[3]{27} = 3$$

Hence, the value of a is 3.

6.

Ans: (b) 0

Explanation:

We have

$$p(x) = x^3 + 3x^2 + 3x + 1$$

By the **Remainder Theorem**, if a polynomial $p(x)$ is divided by $(x + a)$, then the remainder is

$$p(-a)$$

Here, the divisor is $(x + 1)$.

So, the remainder is $p(-1)$.

$$\begin{aligned} p(-1) &= (-1)^3 + 3(-1)^2 + 3(-1) + 1 \\ &= -1 + 3 - 3 + 1 \\ &= 0 \end{aligned}$$

Hence, the remainder is 0.

Thus, option (b) is the correct answer.

7.

Ans:

We have

$$p(x) = x^3 - ax^2 + bx + 3$$

Case I:

If $p(x)$ is divided by $(x + 2)$, the remainder is -19 .

By the **Remainder Theorem**,

$$p(-2) = -19$$

Substituting $x = -2$,

$$(-2)^3 - a(-2)^2 + b(-2) + 3 = -19$$

$$-8 - 4a - 2b + 3 = -19$$

$$-5 - 4a - 2b = -19$$

$$4a + 2b = 14$$

Dividing by 2,

$$2a + b = 7 \quad \text{..... (1)}$$

Case II:

If $p(x)$ is divided by $(x - 2)$, the remainder is 17.

$$p(2) = 17$$

Substituting $x = 2$,

$$2^3 - a(2)^2 + b(2) + 3 = 17$$

$$8 - 4a + 2b + 3 = 17$$

$$11 - 4a + 2b = 17$$

$$-4a + 2b = 6$$

Dividing by 2,

$$-2a + b = 3 \quad \dots\dots (2)$$

Solving (1) and (2):

$$2a + b = 7$$

$$-2a + b = 3$$

Subtracting,

$$4a = 4 \Rightarrow a = 1$$

Substituting $a = 1$ in (1),

$$2(1) + b = 7$$

$$b = 5$$

$$a + b = 1 + 5 = 6$$

Hence, proved that $a + b = 6$.

8.

Ans:

Let

$$P(x) = x^3 + 2x^2 - 5ax - 8$$

Since $P(x)$ is divided by $(x - 2)$, by the **Remainder Theorem**,

$$p = P(2)$$

$$p = 2^3 + 2(2)^2 - 5a(2) - 8$$

$$p = 8 + 8 - 10a - 8$$

$$p = 8 - 10a \quad \dots\dots (1)$$

Now let

$$Q(x) = x^3 + ax^2 - 12x - 6$$

Since $Q(x)$ is divided by $(x - 3)$,

$$q = Q(3)$$

$$q = 3^3 + a(3)^2 - 12(3) - 6$$

$$q = 27 + 9a - 36 - 6$$

$$q = 9a - 15 \quad \dots\dots (2)$$

Given,

$$q - p = 10$$

Substituting from (1) and (2),

$$(9a - 15) - (8 - 10a) = 10$$

$$9a - 15 - 8 + 10a = 10$$

$$19a - 23 = 10$$

$$19a = 33$$

$$a = \frac{33}{19}$$

Hence, the value of a is

$$\boxed{\frac{33}{19}}$$

9.

Ans:

We have

$$p(x) = x^3 - 2x^2 + ax + 12$$

By the **Remainder Theorem**, if a polynomial $p(x)$ is divided by $(x + 1)$, then the remainder is

$$p(-1)$$

Given remainder = 20, so

$$p(-1) = 20$$

Substituting $x = -1$ in $p(x)$,

$$(-1)^3 - 2(-1)^2 + a(-1) + 12 = 20$$

$$-1 - 2 - a + 12 = 20$$

$$9 - a = 20$$

$$-a = 11$$

$$a = -11$$

Hence, the value of a is -11 .

Thus, option (d) is the correct answer.

10.

Ans: (b) 38

Explanation:

We have

$$p(x) = 2x^2 - 7x - 1$$

By the **Remainder Theorem**, if a polynomial $p(x)$ is divided by $(x + 1)$, then the remainder is

$$p(-a)$$

Here, the divisor is $(x + 3)$.

So, the remainder is $p(-3)$.

Substituting $x = -3$,

$$\begin{aligned} p(-3) &= 2(-3)^2 - 7(-3) - 1 \\ &= 2(9) + 21 - 1 \\ &= 18 + 21 - 1 \\ &= 38 \end{aligned}$$

Hence, the remainder is 38.

Thus, option (b) is the correct answer.

11.

Ans: (d) 10

Explanation:

Let

$$p(x) = x^3 + 2x^2 - 13x + k$$

Since $(x - 2)$ is a factor of $p(x)$, by the **Factor Theorem**,

$$p(2) = 0$$

Substituting $x = 2$,

$$2^3 + 2(2)^2 - 13(2) + k = 0$$

$$8 + 8 - 26 + k = 0$$

$$16 - 26 + k = 0$$

$$k = 10$$

Thus, option (d) is the correct answer.